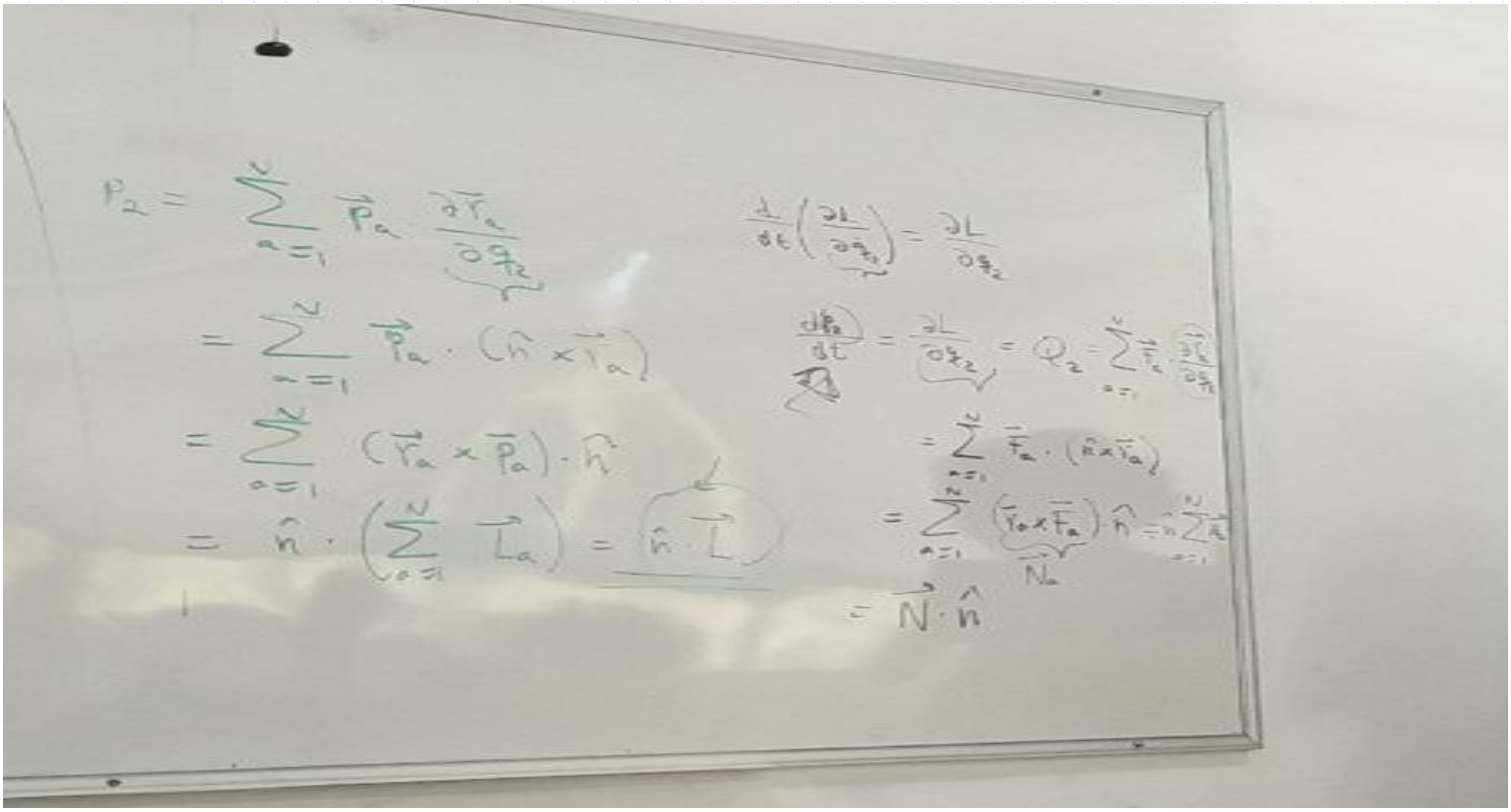
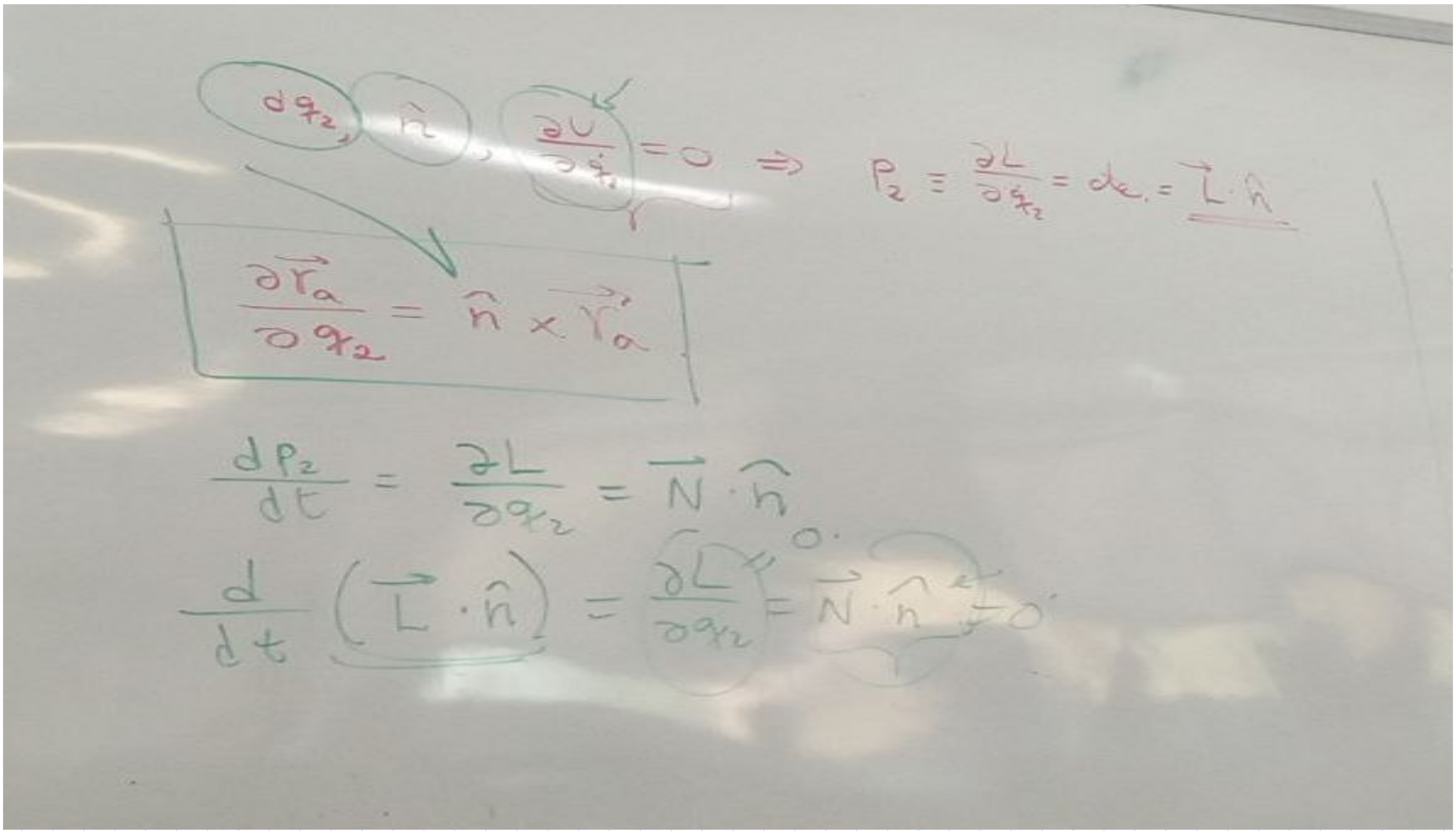
Mecánica teórica

Clase # 21

18/09/2025



Tenemos

$\vec{r} = x\,\hat{x}$

$$L = \frac{m}{2}\dot{x}^2 \quad ; \quad \frac{\partial L}{\partial \dot{x}} = m\dot{x} = p_x$$

$$\boxed{p_x \equiv \frac{\partial L}{\partial \dot{x}}} \longleftrightarrow \boxed{p_i \equiv \frac{\partial L}{\partial \dot{q}_i}}$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} \quad ; \quad \dot{x}\frac{\partial L}{\partial \dot{x}} = m\dot{x}^2 = 2T$$

$$\dot{x}\frac{\partial L}{\partial \dot{x}} = 2L$$

$$\Longrightarrow \boxed{\dot{x}\frac{\partial L}{\partial \dot{x}} - L = L}$$

Función $h$

$$\boxed{h = \sum_{i=1}^{n} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L}$$

$\rightarrow$ Transformada de Legendre

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2$$

$$h = \dot{x}\frac{\partial L}{\partial \dot{x}} - L$$

$\vdots$

$$= \dot{x}\, m\dot{x} - \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m w^2 x^2$$

$$h = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m w^2 x^2 = T + U = E, = \vec{N}\cdot\hat{n}$$

Para particula libre

$$P.L. \} \quad h = E = \frac{1}{2}m\dot{x}^2$$

$$\frac{dh}{dt} = \dot{E} = \frac{1}{2}m\,2\dot{x}\ddot{x} = P\ddot{x} = 0$$

$$\frac{dh}{dt} = \frac{d}{dt}\left[\frac{1}{2}m\dot{x}^2 + \frac{1}{2}m w^2 x^2\right] = \frac{1}{2}m\,2\dot{x}\ddot{x} + \frac{1}{2}m w^2\,2x\dot{x}$$

Tenemos

Factorizando ...

$$= m\dot{x}\left[\ddot{x} + w^2 x\right] = 0$$

Ec. de movimiento para un oscilador armónico no amortiguado ni forzado

$$\boxed{\ddot{x} + w^2 x = 0}$$

$$\ddot{x} = 0$$

Ec. de movimiento para partícula libre.

Tenemos

$$\frac{dh}{dt} = \frac{d}{dt}\sum_{i=1}^{n} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - \frac{dL}{dt}$$

$$\vdots$$

$$= \sum_{i=1}^{n} \frac{d}{dt}\left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i}\right) - \frac{d}{dt}$$

Distribuyendo la derivada.

$$\sum_{i=1}^{n} \left[\underbrace{\frac{d}{dt}(\dot{q}_i)}_{\ddot{q}_i} \cdot \frac{\partial L}{\partial \dot{q}_i} + \dot{q}_i \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\right] - \frac{d}{dt}$$

Tenemos ...

$$= \sum_{i=1}^{n}\left[\ddot{q}_i \frac{\partial L}{\partial \dot{q}_i} + \dot{q}_i \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\right] - \sum_{i=1}^{n}\left[\frac{\partial L}{\partial q_i}\dot{q}_i + \frac{\partial L}{\partial \dot{q}_i}\ddot{q}_i\right] - \frac{\partial L}{\partial t}$$

Dependencia de la Lagrangiana

$(q, \dot{q}, t)$